

PRODUCT REQUIREMENTS DOCUMENT (PRD)

CivicPulse

Public Service Workflow Platform

Version: 1.0

Status: Draft

Owner: Product Team

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1. Executive Summary

CivicPulse adalah platform digital berbasis web yang memungkinkan masyarakat melaporkan permasalahan fasilitas publik serta memungkinkan instansi pemerintah memproses, memantau, dan menyelesaikan laporan tersebut secara transparan, terstruktur, dan terukur.

Berbeda dengan aplikasi pelaporan konvensional, CivicPulse dirancang sebagai Public Service Workflow Platform yang mengelola siklus hidup laporan dari pelaporan hingga penyelesaian melalui workflow operasional yang terdokumentasi, dashboard analitik, audit trail, dan sistem notifikasi.

2. Problem Statement

Banyak pemerintah daerah menghadapi masalah:

- Laporan masyarakat tersebar di berbagai kanal.
- Tidak ada standar proses penanganan.
- Sulit melacak progres laporan.
- Kurangnya transparansi terhadap masyarakat.
- Sulit mengukur performa instansi.
- Data laporan tidak terpusat.

Akibatnya:

- Respons lambat.
 - Kepercayaan publik menurun.
 - Sulit melakukan evaluasi kinerja.
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3. Product Vision

Menjadi platform pelaporan dan manajemen layanan publik yang transparan, terukur, dan mudah digunakan oleh masyarakat maupun instansi pemerintah.

4. Business Goals

Short-Term

- Digitalisasi proses pelaporan masyarakat.
- Mempermudah proses tindak lanjut laporan.
- Menyediakan dashboard monitoring realtime.

Mid-Term

- Menjadi platform standar pelaporan pemerintah daerah.
- Mengurangi waktu respons laporan.

Long-Term

- Menjadi ekosistem Smart City Service Management Platform.
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5. Product Goals

- Mempermudah masyarakat membuat laporan.
 - Mempercepat penanganan laporan.
 - Menyediakan transparansi proses.
 - Menyediakan data analitik untuk pengambilan keputusan.
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6. User Personas

Citizen

Karakteristik:

- Masyarakat umum
- Pengguna smartphone
- Tidak memiliki pengetahuan teknis

Tujuan:

- Melaporkan masalah
- Memantau status laporan

Pain Point:

- Tidak tahu progres laporan
 - Tidak tahu instansi yang menangani
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Government Officer

Karakteristik:

- Pegawai instansi pemerintah
- Mengelola laporan masyarakat

Tujuan:

- Memproses laporan
- Memberikan tindak lanjut

Pain Point:

- Laporan tidak terorganisir
 - Sulit memonitor workload
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Super Admin

Karakteristik:

- Administrator sistem

Tujuan:

- Mengelola instansi
 - Mengelola user
 - Mengawasi performa sistem
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7. Success Metrics

Business KPI

- Total laporan per bulan
- Tingkat penyelesaian laporan
- Average Response Time
- Average Resolution Time

Product KPI

- Daily Active Users

- Monthly Active Users
- Report Completion Rate
- Dashboard Engagement Rate

Operational KPI

- SLA Compliance
 - Escalated Reports
 - Open Reports
 - Closed Reports
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8. User Journey

Citizen Login



Create Report



Report Submitted



Assigned To Agency



In Progress



Resolved



Closed



Citizen Satisfaction

9. Core Features

Citizen Features

Authentication

- Register
- Login
- Logout
- Reset Password

Report Management

- Create Report
- Upload Evidence
- View Report History
- Track Status

Dashboard

- Personal Reports
 - Report Statistics
 - Notifications
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Officer Features

Dashboard

- Assigned Reports
- Pending Reports
- Urgent Reports

Report Processing

- Update Status
- Add Notes
- Upload Resolution Evidence

Export

- CSV Export
 - PDF Export
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Super Admin Features

User Management

- Create User

- Disable User
- Assign Role

Agency Management

- Create Agency
- Edit Agency
- Delete Agency

System Monitoring

- Activity Monitoring
- Audit Log Monitoring

10. Workflow Management

Report Lifecycle

DRAFT

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SUBMITTED

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ASSIGNED

↓

IN_PROGRESS

↓

RESOLVED

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CLOSED

Status Definitions

SUBMITTED

Laporan berhasil dibuat.

ASSIGNED

Laporan telah diteruskan ke instansi terkait.

IN_PROGRESS

Instansi sedang menangani.

RESOLVED

Masalah telah diperbaiki.

CLOSED

Laporan selesai diverifikasi.

11. Functional Requirements

FR-001

Citizen dapat membuat laporan.

FR-002

Citizen dapat mengunggah foto.

FR-003

Citizen dapat melihat riwayat laporan.

FR-004

Officer dapat mengubah status laporan.

FR-005

Officer dapat memberikan tanggapan.

FR-006

Officer dapat mengunggah bukti penyelesaian.

FR-007

System mengirim notifikasi saat status berubah.

FR-008

System mencatat seluruh aktivitas pengguna.

FR-009

System menghasilkan dashboard analitik.

FR-010

Admin dapat mengelola instansi.

12. Non Functional Requirements

Security

- JWT Authentication
- Role Based Access Control
- Session Validation
- CSRF Protection
- Rate Limiting

Performance

- First Load < 3 Seconds
- API Response < 500ms

Scalability

- 100.000+ Reports
- 10.000+ Users

Availability

- 99.9% Uptime
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13. RBAC Design

Citizen

Permissions:

- Create Report
- View Own Report

Officer

Permissions:

- View Assigned Reports
- Update Status
- Add Resolution

Super Admin

Permissions:

- Full Access
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14. Database Design

Core Tables

Users

Agencies

Reports

ReportAttachments

ReportComments

ReportHistory

Notifications

AuditLogs

Sessions

15. Geolocation Design

Each report stores:

- Latitude
- Longitude
- Address
- District
- City
- Province

Features:

- Map View
 - Area Filter
 - Geographic Analytics
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16. Notification System

Channels

- In-App Notification
- Email Notification

Events

- Report Created
 - Report Assigned
 - Report Updated
 - Report Resolved
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17. Audit Trail

Track:

- Login Activity
- Report Changes
- Status Changes
- User Actions

Data:

- Actor
 - Action
 - Timestamp
 - Resource
 - Before Value
 - After Value
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18. Analytics Dashboard

Public Dashboard

- Total Reports
- Resolved Reports
- Resolution Rate

- Average Handling Time

Admin Dashboard

- Reports By Category
 - Reports By District
 - Agency Performance
 - Monthly Trends
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19. System Architecture

Architecture Style:

Modular Monolith

Layers:

Presentation Layer

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Application Layer

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Domain Layer

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Infrastructure Layer

20. Recommended Tech Stack

Frontend

- Next.js 16
- TypeScript
- Tailwind CSS
- shadcn/ui
- TanStack Query
- Zustand

Backend

- Next.js Route Handlers
- Service Layer

- Repository Layer

Database

- PostgreSQL
- Prisma ORM

Authentication

- Better Auth

Storage

- MinIO
- S3 Compatible Storage

Queue

- Redis
- BullMQ

Monitoring

- Sentry
- OpenTelemetry

Deployment

- Docker
- Coolify
- VPS

21. Security Architecture

- Middleware Authorization
- RBAC Enforcement
- Server Side Validation
- Zod Validation
- Password Hashing
- Audit Logging

22. MVP Scope

Included

- Authentication
- Report Submission
- Attachment Upload

- Workflow Status
- Dashboard
- Notifications
- Analytics

Excluded

- AI Classification
 - Smart Routing
 - Mobile App
 - WhatsApp Integration
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23. Future Roadmap

V1

Citizen Reporting Platform

V2

Multi Agency Workflow

V3

AI Categorization

V4

Smart City Analytics

V5

Predictive Infrastructure Monitoring

24. Risks

Technical Risks

- Large File Upload
- Database Growth
- Notification Delivery

Business Risks

- Low User Adoption
- Poor Agency Response

Operational Risks

- Human Error
 - Misclassification
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25. Definition of Success

Produk dianggap berhasil apabila:

- 80% laporan dapat diproses secara digital.
- Resolution Rate > 70%.
- Average Response Time turun 50%.
- Dashboard digunakan aktif oleh instansi.
- Citizen Satisfaction meningkat secara konsisten.